National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Probability & Random processes	Course Code:	EE2011
	Program:	BS Electrical Engineering	Semester:	Spring 2025
	Assignment Submission Date:	12-May-2025	Total Marks:	100
			Weight:	3
	Section:	BEE-4B	Page(s):	3
	Exam Type:	Assignment- 3	CLO #	2

Student Name:______________________________ Roll No.________________ Section:_________

Instructions:
1. Submission should be on a hardcopy and handwritten
2. Solve on A4 sheets.

Question No. 1 (CLO No. 2) Marks: 10

According to a survey by the Administrative Management Society, one-half of U.S. companies give employees 4 weeks of vacation after they have been with the company for 15 years. **Analyze** and find the probability that among 6 companies surveyed at random, the number that give employees 4 weeks of vacation after 15 years of employment is

(a) Anywhere from 2 to 5;

(b) Fewer than 3.

Question No. 2 (CLO No. 2) Marks: 10

It is conjectured that an impurity exists in 30% of all drinking wells in a certain rural community. In order to gain some insight into the true extent of the problem, it is determined that some testing is necessary. It is too expensive to test all of the wells in the area, so 10 are randomly selected for testing. **Analyze**

(a) Using the binomial distribution, what is the probability that exactly 3 wells have the impurity, assuming that the conjecture is correct?

(b) What is the probability that more than 3 wells are impure?

Question No. 3 (CLO No. 2) Marks: 10

In testing a certain kind of truck tire over rugged terrain, it is found that 25% of the trucks fail to complete the test run without a blowout. Of the next 15 trucks tested, **Analyze and** find the probability that

(a) From 3 to 6 have blowouts;

(b) Fewer than 4 have blowouts;

(c) More than 5 have blowouts.

Question No. 4 (CLO No. 2) Marks: 10

Assuming that 6 in 10 automobile accidents are due mainly to a speed violation, **Analyze and** find the probability that among 8 automobile accidents, 6 will be due mainly to a speed violation

(a) By using the formula for the binomial distribution;

(b) By using Table A.1

Question No. 5 (CLO No. 2) **Marks: 10**

From a lot of 10 missiles, 4 are selected at random and fired. If the lot contains 3 defective missiles that will not fire, **Analyze** what is the probability that

(a) All 4 will fire?

(b) At most 2 will not fire?

Question No. 6 (CLO No. 2) **Marks: 10**

A government task force suspects that some manufacturing companies are in violation of federal pollution regulations with regard to dumping a certain type of product. Twenty firms are under suspicion but not all can be inspected. Suppose that 3 of the firms are in violation. **Analyze**

(a) What is the probability that inspection of 5 firms will find no violations?

(b) What is the probability that the plan above will find two violations?

Question No. 7 (CLO No. 2) **Marks: 10**

On average, a textbook author makes two word processing errors per page on the first draft of her text book. **Analyze** What is the probability that on the next page she will make

(a) 4 or more errors?

(b) No errors?

Question No. 8 (CLO No. 2) **Marks: 10**

The daily amount of coffee, in liters, dispensed by a machine located in an airport lobby is a random variable X having a continuous uniform distribution with A = 7 and B = 10. **Analyze and** Find the probability that on a given day the amount of coffee dispensed by this machine will be

(a) At most 8.8 liters;

(b) More than 7.4 liters but less than 9.5 liters;

(c) At least 8.5 liters

Question No. 9 (CLO No. 2) **Marks: 10**

A research scientist reports that mice will live an average of 40 months when their diets are sharply restricted and then enriched with vitamins and proteins. Assuming that the lifetimes of such mice are normally distributed with a standard deviation of 6.3 months, **Analyze and** find the probability that a given mouse will live

(a) More than 32 months;

(b) Less than 28 months;

(c) Between 37 and 49 months.

Question No. 10 (CLO No. 2) **Marks: 10**

The weights of a large number of miniature poodles are approximately normally distributed with a mean of 8 kilograms and a standard deviation of 0.9 kilogram. If measurements are recorded to the nearest tenth of a kilogram, **Analyze and** find the fraction of these poodles with weights

(a) Over 9.5 kilograms;

(b) Of at most 8.6 kilograms;

(c) Between 7.3 and 9.1 kilograms inclusive.